

Project Presentation Rubric and Guidelines

Programming and Data Structures — Week 15

Dr. Innocent Nyalala

Purpose

The course project asks each student (or pair) to design, implement, and present a real application that combines at least two distinct modules from this course. This document specifies exactly what is expected and how it will be graded.

Requirements

1. **Combine at least two course modules.** For example: a route planner using Weeks 11-13's graph algorithms plus Week 8's priority queues; a text search tool using Week 8's hash tables plus Week 9's dynamic programming for fuzzy matching; a scheduling tool using Week 10's greedy algorithms plus Week 3's queues.
2. **A working demonstration.** The application must actually run, either live during the presentation or as a recorded screen capture, on real or realistic input data — not solely a single hand-picked toy example.
3. **A measured performance analysis.** Any claim about speed, efficiency, or scalability must be backed by an actual measurement (timing code, a profiler, or a clear before/after comparison), not an assumption. This is the same “measure, don't assume” discipline applied throughout this course's own lecture construction.
4. **A written or slide-based explanation** of the problem, the technique(s) chosen, the alternatives considered and why they were not chosen, and the trade-offs made.

Rubric

Component	Weight	What is assessed
Problem framing	15%	Is the problem clearly stated and genuinely non-trivial?

Component	Weight	What is assessed
Technique justification	25%	Is the chosen data structure/algorithm explained and compared against a real alternative?
Implementation and demonstration	30%	Does it run correctly on realistic input, not only a toy example?
Performance analysis	20%	Is any efficiency claim backed by an actual measurement?
Q&A and defense of trade-offs	10%	Can the presenter explain what they would do differently, and defend the choices made under questioning?

Common pitfalls to avoid

- Choosing a technique because it was the most recently covered lecture, rather than because it fits the problem. Apply the tool-selection reasoning process from the Week 15 course review lecture.
- Claiming a performance improvement without measuring the baseline being compared against.
- Presenting a working demo with no discussion of what would break, or slow disproportionately, at a substantially larger scale than shown.
- Combining two modules superficially (e.g., using an array somewhere and calling it “combining two modules”) rather than genuinely relying on both techniques for the application to function.

Presentation logistics

- Length: 10 minutes presentation, 5 minutes questions.
- Slides or a live notebook/demo are both acceptable; a working, runnable artifact is required either way.
- Submit source code and any written report to the course project repository before your presentation slot.